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crystALL: Purely data-driven automated structure prediction for unidentified chemical compounds

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Summary

- (2 sentences) Motivation for materials discovery
  - CALPHAD (Olson & Liu, 2023)
  - Going beyond stability. It is not enough (Zunger, 2019) because of the complexities involved in the synthesis of new materials. Phase coexistance can take properties to extreme, e.g. infinitely negative thermal expansion (Liu et al., 2022).
- Typically done through combinatoric search based on protoypes and DFT
  - Proposing new structures for compounds is not a new idea based on geometrical constraints (Bail, 2005)
  - AFLOW Database (Toher et al., 2018) built by expert-intuition driven population of AFLOW Prototype Library (Mehl et al., 2016) with new chemistries in a systematic way
  - Similarity driven prediction of substitutions (Wang et al., 2021)
  - Compositional complexity grows quickly with dimensionality, considering thhat substitutions (A. M. Krajewski et al., 2024)
- (3 sentences) Past deployment on the Nd-Bi system (Im et al., 2022) and Figure 7a of (Im et al., 2022) showing excellent ordering of the candidates when evaluated vs different DFT methods. Past deployment in Al-Fe (Shang et al., 2021).

Statement of need

(statement of need)

- Motivation slightly reiterating the summary with more focus on impact and need
- (**Rushi**, 3 sentences) *(Importance of Ru-Zn in the study of catalysts. Importance of KCl-LaCl3 system in the study of molten salts. How `crystALL` is continuously used in the lab to predict structures of unknown compounds and if the automated experience has been positive.)* To enhance nylon manufacturing efficiency, Ru-based catalysts are recognized as highly effective catalysts in synthesizing the key chemical intermediate through the benzene partial hydrogenation reaction (Zhang et al., 2013). Achieving precise control over this process is important to ensure optimal outcomes. Improvement of hydrogenation selectivity involves designing diverse active site ensembles in intermetallic catalysts (Dasgupta et al., 2022). Using crystALL, we have explored over 165, 000 configurations of Ru-Zn intermetallic compounds, including Ru_3Zn, RuZn, Ru_4Zn_9, $RuZn_3$, Ru_3Zn_{10}, $RuZn_6$, $RuZn_{10}$, and $RuZn_{13}$. We are currently examining their structures and stability through pySIPFENN (A. M. Krajewski et al., 2022), while simultaneously evaluating active site ensembles to design novel catalysts with high activity and selectivity. In the journey towards clean energy and carbon footprint reduction, the

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development of Molten Salt Reactors (MSRs) is essential, which requires high reliability and efficiency. crystALL serves as a valuable tool for identifying structures for complex molten salts, particularly in cases of limited or conflicting experimental data. For example, crystALL facilitated the prediction of structures for KLa_3Cl_{10} and $K_3La_5Cl_{18}$ compounds in the KCl-$LaCl_3$ system, thereby validating their stability reported in various literature sources (Seifert et al., 1985) (Song & Zheng, 1995).

- (**Adam**, 1 sentence) Conclusion on the importance of automation for user tool.

Methods

MPDD Database

Contains: - OQMD (Shen et al., 2022) - AFLOW Database (Toher et al., 2018) - Materials Project (Jain et al., 2013) - NIST-JARVIS (Choudhary et al., 2020) - COD (Gražulis et al., 2019) - GNoME (Merchant et al., 2023)

API: - native pymongo - OPTIMADE (Evans et al., 2024)

- use of `pymatgen` (Ong et al., 2013) for data processing

Predictive Modeling Pipeline

(about the pySIPFENN, MPDD-ALIGNN, and other models)

- pySIPFENN software (A. M. Krajewski et al., 2022) and models on Zenodo (A. Krajewski et al., 2020)
- ALIGNN (Choudhary & DeCost, 2021)

(schematic of crystALL software)

Compound-Based Structure Prediction

(about the structure prediction)

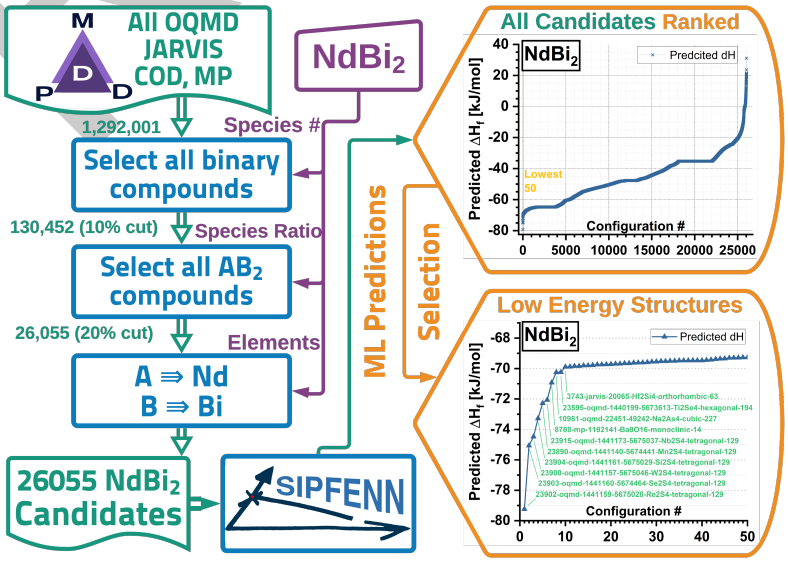


Figure 1: Compound-Based Structure Search. Reproduced from (Im et al., 2022) under permission.

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Composition-Based Structure Prediction

(uncertainty in compound composition extends the previous method)

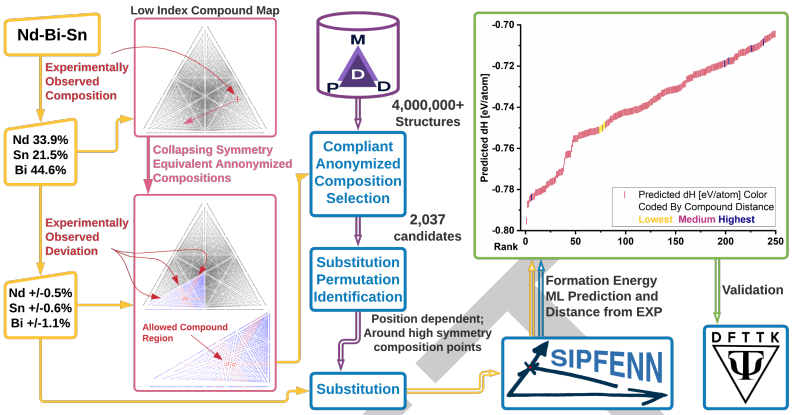


Figure 2: Composition-Based Structure Search.

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- All Phases Research Lab members who provided feedback on the software

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